

PEFT-Arena

Official Code Release

PEFT-Arena: Understanding Parameter-Efficient Finetuning from a Stability-Plasticity Perspective

Evaluate PEFT through the joint lens of target adaptation, retained general ability, and update geometry.

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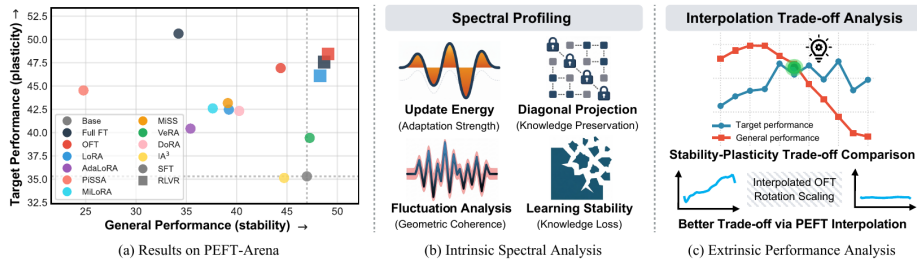
Paper Poster Code Citation

2 Target Domains

3 Retention Benchmarks

2 Training Regimes

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PEFT-Arena combines benchmark evaluation, spectral retention-adaptation profiling, and interpolation-based trade-off analysis to study how PEFT methods balance plasticity and stability.

Paper Summary

Abstract

Parameter-efficient finetuning (PEFT) has become the standard approach for adapting large foundation models, yet existing evaluations focus almost exclusively on downstream accuracy while largely overlooking the preservation of pretrained capabilities. PEFT-Arena argues that PEFT should be evaluated through the lens of the stability-plasticity dilemma: the trade-off between task adaptation and resistance to forgetting.

The benchmark jointly measures target-domain performance on mathematical and medical reasoning together with retained general ability on BBH, IFEval, and NQ, under both supervised finetuning (SFT) and reinforcement learning with verifiable rewards (RLVR / GRPO). Across Qwen2.5-7B and Llama3.2-3B-Instruct, all PEFT methods exhibit distinct stability-plasticity trade-off patterns, with OFT typically defining the most favorable Pareto frontier at comparable parameter budgets.

Beyond benchmark-level metrics, the paper introduces a spectral retention-adaptation profiling framework that decomposes weight updates into retention and adaptation components and measures their smoothness. Smoother spectral profiles correlate with better retention. The paper further identifies an overshoot phenomenon in SFT and studies interpolation-based trade-off recovery, including the geometry-aware iOFT variant.

Driving Questions

Research Questions

RQ1

Which PEFT method achieves the best stability-plasticity trade-off?

Benchmark PEFT by target adaptation and retained general capability across math and medical reasoning, under both SFT and RLVR.

RQ2

What internal mechanisms govern forgetting and retention?

Analyze spectral geometry and interpolation trajectories to understand why some parameterizations stay stable while others overshoot.

Benchmark Design

Protocol

Target Domains

- **Math:** Math-500, AMC23, AIME24

- **Medical:** a curated suite including MedMCQA, MedQA, PubMedQA, MMLU-Pro, GPQA (Medical), Lancet/NEJM/MedBullets-style problems, and MedXpertQA

General Retention

- **IFEval** for instruction following
- **NQ** for knowledge and understanding
- **BBH** for broad reasoning retention

Models and Training

- **Backbones:** Qwen2.5-7B and Llama3.2-3B-Instruct
- **Regimes:** supervised finetuning and RLVR with GRPO
- **Target data:** OpenR1-Math and MedThink

Compared Methods

- **Full FT**
- **LoRA family:** LoRA, AdaLoRA, DoRA, MiSS, VeRA, PiSSA, MiLoRA, KeepLoRA
- **Other PEFT:** OFT and IA³

Main Findings

What PEFT-Arena Shows

SFT exposes a sharp trade-off

Full finetuning tends to deliver the strongest target gains, but it also incurs the largest forgetting. Target-only reporting systematically overestimates post-training quality.

OFT sits on the favorable frontier

At comparable parameter budgets, OFT consistently balances adaptation and retention better than additive low-rank baselines across the paper’s main settings.

RLVR is more forgetting-resistant

Under on-policy RLVR with GRPO, models often achieve stable target improvements with substantially less general-capability erosion than under SFT.

Geometry matters

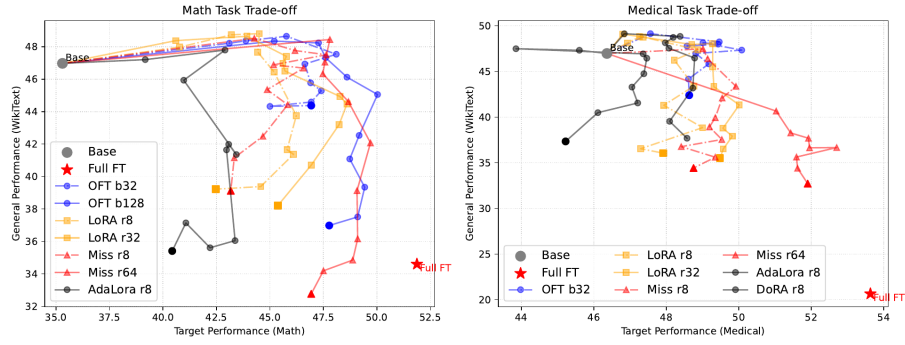
Smoother spectral retention and adaptation profiles correlate with better retention. Interpolation also reveals an SFT overshoot region and motivates geometry-aware trade-off recovery via iOFT.

Results

Representative Figures

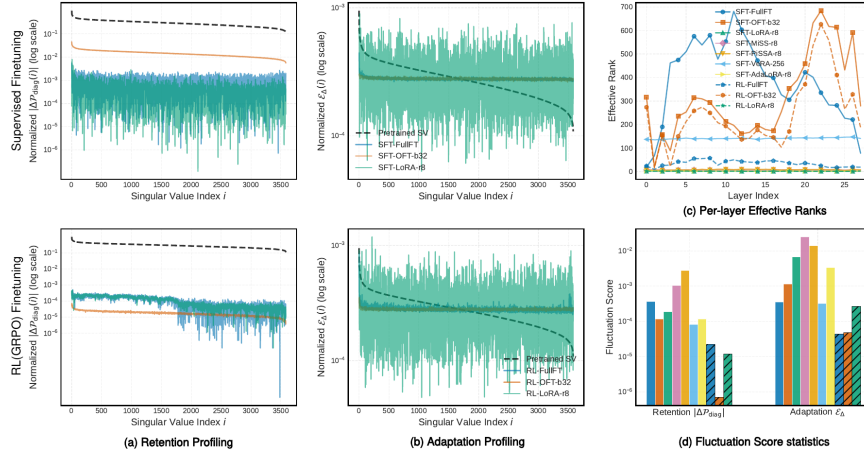
Stability-Plasticity Benchmark

The benchmark reports target-domain performance together with retained general capability, rather than reducing post-training quality to a single downstream score. The strongest PEFT methods are those that move toward the upper-right frontier rather than simply maximizing target performance.



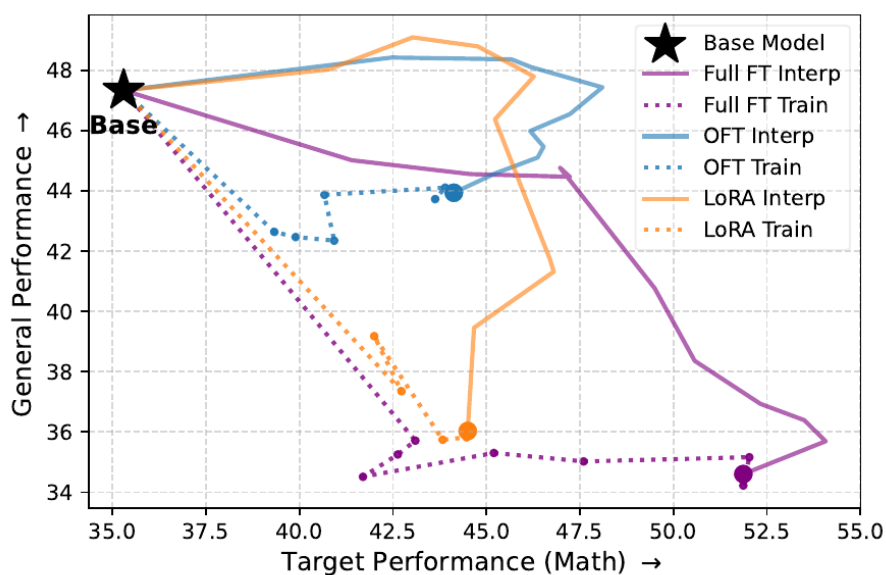
Spectral Retention-Adaptation Profiling

The paper decomposes updates into retention and adaptation views and quantifies local fluctuation across singular directions. OFT tends to remain in a smoother and more coherent spectral regime than spikier additive updates.



Interpolation and Overshoot

Training trajectories and interpolation trajectories are misaligned. This reveals that final SFT checkpoints often overshoot the best trade-off point, and motivates interpolation-based recovery and iOFT.



Poster

Project Poster

The poster summarizes the benchmark protocol, the main stability-plasticity findings, the spectral mechanism analysis, and the interpolation/iOFT story in a single overview.

[Open Poster PDF](#)

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Research Questions

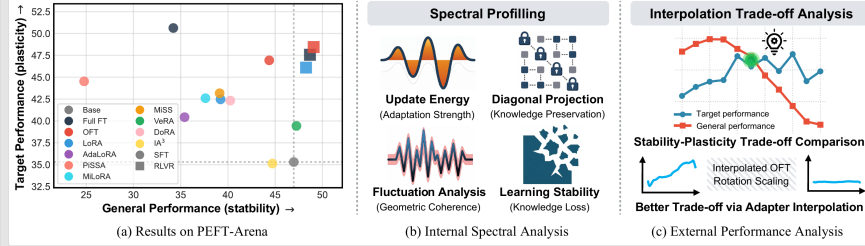
RQ1

Which PEFT method achieves the best stability-plasticity trade-off?
Benchmark PEFT by target adaptation and general capability retention across math and medical reasoning.

RQ2

What internal mechanisms govern the stability-plasticity trade-off?
Probe PEFT through spectral geometry, then use Interpolation to turn that understanding into better operating points.

Overview



Part I. Benchmark

Evaluate target gains together with retained general capability.

Part II. Internal Mechanism

Spectral profiling reveals why some updates forget less.

Part III. Trade-off by Interpolation

Overshoot motivates interpolation and the geometry-aware iOFT.

Part I. PEFT-Arena Benchmark

We evaluate both **Owen2.5-7B-base** and **Llama3.2-3B-Instruct** under supervised fine-tuning (SFT) and **GRPO-based reinforcement learning**, measuring both target and general performance.

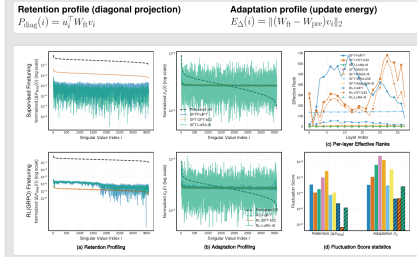
Method	H.P.	Q Tr.	Owen2.5-7B-base					Llama3.2-3B-Instruct				
			Math	T	Med	T	Med	Math	T	Med	T	Med
SFT												
Base	—	7.61B	35.30	46.97	46.36	46.97	3.21B	27.63	53.03	41.44	56.76	
Full FT	—	6.53B	50.63	34.22	53.63	34.41	2.85B	33.90	39.83	44.26	26.03	
OFT	b16	8.49M	42.33	42.58	46.17	45.09	7.08M	29.43	41.08	39.22	40.97	
	b32	17.55M	46.93	44.37	48.63	42.40	11.55M	30.60	40.73	39.50	40.50	
OFT	b64	35.68M	46.23	35.97	49.47	39.11	24.97M	29.30	39.75	40.77	37.70	
OFT	b128	71.92M	47.77	36.98	52.40	36.88	47.34M	32.23	36.26	42.17	34.26	
LoRA	r1a8	10.09M	42.33	41.66	47.12	36.42	6.90M	24.30	35.79	36.92	31.84	
LoRA	r1a16	20.19M	42.47	39.22	47.91	36.06	12.16M	24.07	36.57	38.34	27.99	
LoRA	r1a32	40.37M	44.87	34.91	47.86	34.86	24.31M	24.97	37.55	39.21	29.18	
LoRA	r1a64	80.74M	45.37	38.21	49.48	35.50	48.63M	25.90	37.20	39.33	30.69	
AdaLoRA	r1a16	30.28M	40.43	35.41	45.22	37.34	18.24M	20.83	34.53	37.11	36.29	
PISSA	r1a16	20.19M	44.53	24.78	26.16	18.05	12.16M	0.67	7.08	21.17	9.75	
MLoRA	r1a16	20.19M	42.60	37.62	46.83	35.88	12.16M	23.60	35.59	37.64	29.23	
MISS	r8	11.12M	43.17	39.12	48.75	34.43	6.19M	23.37	33.93	40.16	31.71	
MISS	r16	89.00M	46.93	32.77	51.90	32.72	49.55M	28.63	34.96	41.96	27.07	
VeRA	r256	1.44M	39.43	47.25	28.50	47.01	0.82M	28.80	46.79	40.68	48.94	
DoRA	r1a16	21.58M	42.33	40.25	48.04	36.06	12.93M	23.83	35.65	38.25	27.53	
IA ¹	—	1.82M	35.13	44.71	29.70	45.72	0.92M	29.70	45.72	39.13	45.87	
RLVR with GRPO												
Full FT	—	6.53B	47.57	48.68	46.24	43.22	2.85B	29.80	52.20	45.88	51.81	
OFT	b32	8.49M	48.37	48.90	46.79	47.24	11.55M	29.97	50.04	44.99	52.31	
LoRA	r1a16	20.19M	46.10	48.27	47.08	42.80	24.31M	28.83	52.17	46.24	53.54	

Takeaways.

- Full FT tends to learn more and forget more, while PEFT behavior depends on both parameter budget and parameterization.
- Within additive LoRA-style PEFT, SVD-space methods do not yield stable performance in SFT.
- OFT, based on orthogonal rotation, achieves the best **stability-plasticity trade-off**.

Part II. Spectral Mechanism

Key idea. Better retention aligns with smoother spectral deviations.

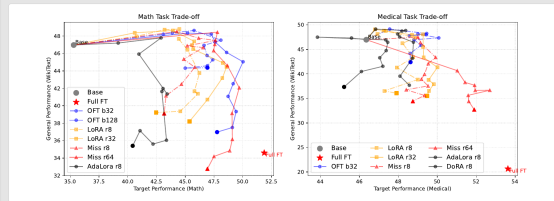


What the profiling shows.

- Additive low-rank PEFT is often spectrally spiky under SFT.
- OFT stays in a smoother, more globally systematic spectral regime via near-orthogonal rotations.
- Smoother retention and adaptation profiles explain why OFT forgets less while staying adaptive.

Part III. Interpolation, Overshoot, and iOFT

Trade-off curves. Interpolating between the base and finetuned checkpoints consistently improves the frontier.

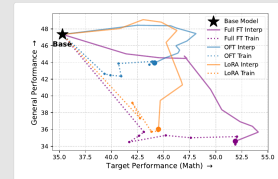


iOFT: geometry-aware OFT adjustment.

- Rescale ($\downarrow$) layer by layer to flatten rotation strength.
- SafeRho**: match the first-five-layer average.
- MinRho**: match the minimum across layers.
- Recover general ability without sweeping a global interpolation coefficient.

Variant	MT	MG	MedT	MedG
OFT	46.93	44.37	48.63	42.40
SafeRho	47.57	46.69	50.01	47.61
MinRho	47.83	46.88	49.76	47.76

The Overshoot Phenomenon. Misalignment between training trajectories and parameter interpolation trajectories inspired our trade-off method.



Method	$\Delta_{\text{pass}@1}$	$\Delta_{\text{pass}@64}$	$\Delta_{\text{p}@64 - p@1}$
LoRA	2.30 $\uparrow$	2.93 $\downarrow$	5.23 $\downarrow$
OFT	0.07 $\downarrow$	1.46 $\downarrow$	1.39 $\downarrow$

RLVR after longer GRPO. OFT keeps high-i behavior much better than LoRA (avoid policy collapse with better diversity).

Official Release

Code and Resources

What the Repository Includes

- SFT and RLVR training entrypoints through a unified `run.py` interface
- Math, medical, and general-retention evaluation pipelines
- Checkpoint preparation and adapter merge utilities
- Spectral analysis and plotting code used for the paper’s figures
- Release-ready setup scripts for the bundled third-party dependencies

Quick Commands

```
bash setup_env.sh
```

```
python run.py train sft ...
```

```
python run.py train rl ...
```

```
python run.py eval ...
```

```
python run.py merge ...
```

Take-home message: PEFT should be evaluated and designed through the joint lens of plasticity, retention, and geometry.

Reference

Citation

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  title={PEFT-Arena: Understanding Parameter-Efficient Finetuning from a Stability-Plasticity Perspective},  
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  year={2026},  
  note={Official code release and project page},  
  url={https://github.com/Sphere-AI-Lab/PEFT-Arena}  
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```

Website template originally adapted from Nerfies, then refined for the PEFT-Arena release.